# Exploring ML Models for Detection of Atmospheric Composition in Exoplanets

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**ABSTRACT**

Inferring the chemical composition of exoplanet atmospheres from observational data — atmospheric retrieval — is central to the search for biosignatures and habitable worlds. Classical Bayesian retrieval methods are computationally prohibitive at the scale demanded by missions such as the James Webb Space Telescope (JWST) and the planned Habitable Worlds Observatory (HWO). In this work, we apply two machine learning models — a per-molecule Random Forest (RF) baseline and a plain 1D CNN (CNN1D, 368,588 parameters) — to the INARA ATMOS dataset of 124,320 synthetic rocky-planet atmosphere simulations, predicting $\log_{10}$ surface volume mixing ratios of 12 atmospheric molecules from CLIMA photochemical-climate profiles. On a held-out test set of 18,648 samples from the full 124,320-sample HPC run, the RF achieves a mean $R^2$ of 0.826 and the CNN1D achieves 0.998; the CNN1D outperforms RF on all 12 molecules. Both models perform inference in milliseconds, compared to hours for Bayesian samplers. We analyse failure modes — particularly the near-constant $H_2O$ target and the structurally unlearnable $NH_3$ — and discuss implications for future deployment on real JWST spectra.

*Keywords: exoplanet atmospheres, atmospheric retrieval, machine learning, 1D CNN, random forest, JWST, biosignatures, INARA ATMOS*

## I. INTRODUCTION

The characterisation of exoplanet atmospheres is one of the most active frontiers in astrophysics. By analysing how starlight is filtered through a planet's atmosphere during a transit — a technique known as transmission spectroscopy — astronomers can determine which molecules are present and in what concentrations. The presence, absence, and co-occurrence of specific molecules (notably $O_2$, $O_3$, $CH_4$, $H_2O$, $CO_2$) are key indicators of habitability and potential biological activity.

Traditional atmospheric retrieval algorithms rely on Bayesian sampling methods such as Markov Chain Monte Carlo (MCMC) or nested sampling (e.g., MultiNest). While accurate, these methods require evaluating a forward radiative transfer model thousands of times and can take hours to days per planet. JWST, which began science operations in 2022, is expected to observe hundreds to thousands of transiting exoplanets. Fast, automated retrieval at population scale is therefore a scientific necessity.

Machine learning approaches have been proposed as a solution: once trained on a large library of synthetic atmospheres, an ML model can produce a retrieval in milliseconds. This work trains and evaluates two models on the INARA ATMOS synthetic dataset [1], benchmarking retrieval accuracy across 12 molecules of scientific interest.

The contributions of this work are:

- Training and evaluation of a per-molecule Random Forest on PCA-compressed CLIMA profiles
- Design and training of a plain 1D CNN (CNN1D, 368,588 parameters) operating directly on the (12 × 101) atmospheric profile tensor — four convolutional blocks with no residual connections, followed by 12 per-molecule output heads
- Systematic evaluation on the full 124,320-sample dataset on the Northeastern HPC cluster, demonstrating CNN1D mean $R^2 = 0.998$ on an 18,648-sample test set
- Analysis of failure modes: near-constant targets ($H_2O$), trivially degenerate targets ($NH_3$), and the effect of dataset scale on model accuracy

**Key result:** The CNN1D achieves mean $R^2 = 0.998$ on the 124K test set — near-perfect retrieval accuracy across all learnable molecules — at millisecond inference time, representing a ~10,000× speedup over Bayesian retrieval methods.

## II. BACKGROUND & RELATED WORK

### A. Atmospheric Retrieval

Classical retrieval couples a parametric forward model (line-by-line radiative transfer) with a Bayesian sampler to produce posterior distributions over atmospheric parameters [2]. The computational cost scales with the number of free parameters; for a 12-parameter retrieval on a JWST-quality spectrum, nested sampling typically requires $10^5$–$10^6$ forward model evaluations.

### B. Machine Learning for Retrieval

Márquez-Neila et al. [3] demonstrated that random forests trained on synthetic spectra can recover atmospheric parameters in seconds, at accuracy competitive with nested sampling for high-SNR spectra. Vasist et al. [4] extended this to normalising flows for full posterior estimation. Gebhard et al. [5] applied flow matching to enable posterior retrieval without MCMC. Our work follows this direction, specifically targeting the INARA ATMOS benchmark dataset designed as a standardised training and evaluation resource for ML retrieval methods.

### C. JWST Context

The first JWST exoplanet transmission spectrum — WASP-39 b [6] — confirmed detections of $CO_2$, $SO_2$, $H_2O$, CO, and K in a single observation, validating the instrument's capability. $SO_2$ was identified as a photochemical product of $H_2O$ and $CO_2$, demonstrating that cross-molecule relationships carry chemical information — precisely the type of signal that a multi-output ML model can exploit.

## III. DATASET

### A. INARA ATMOS

The INARA (Investigating Non-trivial Atmospheres for Retrieval Algorithms) dataset [1] contains synthetic rocky-planet atmosphere simulations generated with the ATMOS photochemical-climate model suite (CLIMA + Photochem). The full archive spans approximately 3.1 million simulations; we use the publicly available subset of 124,320 samples across ten tar archives. Each sample represents one simulated rocky exoplanet atmosphere with randomised input conditions (surface pressure, temperature, molecular fluxes).

### B. Input Features — CLIMA Profiles

Each sample provides a CLIMA atmospheric profile: a tensor of shape (12, 101) representing 12 physical variables measured at 101 altitude levels from the surface (level 0) to the top of the atmosphere (~83 km, level 100). After extraction, the array is transposed from (101, 12) to (12, 101) — channels-first layout for convolutional processing.

| Ch. | Variable | Physical Meaning | Typical Range |
|---|---|---|---|
| 0 | J | Actinic radiative flux | 1–101 |
| 1 | P | Atmospheric pressure | $7.7{\times}10^{-6}$ – 1.19 bar |
| 2 | ALT | Altitude (km) | 0 – 82.9 km |
| 3 | T | Temperature | 41 – 332 K |
| 4 | CONVEC | Convective energy flux | varies |
| 5 | DT | Temperature tendency | varies |
| 6 | TOLD | Previous time-step temperature | varies |
| 7 | $FH_2O$ | Water vapour radiative flux | varies |
| 8 | FSAVE | Saved total radiative flux | varies |
| 9 | $FO_3$ | Ozone radiative flux | varies |
| 10 | TCOOL | Radiative cooling rate | –474 – 779 |
| 11 | THEAT | Radiative heating rate | 0 – 340 |

*Table I. CLIMA input channels. The "altitude axis" (101 levels) is not spectral wavelength — it is a physical altitude profile through the atmosphere.*

### C. Targets — Molecular Abundances

For each sample, the surface-layer (row 0) concentrations of 12 molecules are extracted from the photochemistry model output and transformed to $\log_{10}$ volume mixing ratios. $N_2$ requires special handling — it is absent from the photochem array and is instead parsed from a separate mixing_ratios.dat file, with fallback: $N_2 \approx \max(0, 1 - x(H_2O) - x(O_2))$. A floor of –40 prevents –∞ for undetectable trace species.

| Molecule | Scientific Role | Mean ($\log_{10}$) | Std | Min | Max |
|---|---|---|---|---|---|
| $H_2O$ | Habitability indicator | –1.910 | 0.0002 | –1.910 | –1.907 |
| $CO_2$ | Greenhouse gas | –2.004 | 0.841 | –3.398 | –0.456 |
| $O_2$ | Biosignature | –0.702 | 0.172 | –1.699 | –0.046 |
| $O_3$ | Biosignature / UV shield | –7.780 | 0.122 | –8.588 | –7.406 |
| $CH_4$ | Biosignature | –2.895 | 1.823 | –5.788 | –0.886 |
| $N_2$ | Background carrier gas | –0.112 | 0.045 | –1.057 | –0.014 |
| $N_2O$ | Biosignature | –6.589 | 0.153 | –7.485 | –6.183 |
| CO | Photochemical product | –4.383 | 1.334 | –6.860 | –2.830 |
| $H_2$ | Primordial hydrogen | –5.474 | 2.532 | –7.187 | –0.770 |
| $H_2S$ | Volcanic outgassing | –10.355 | 0.395 | –11.162 | –9.712 |
| $SO_2$ | Volcanic outgassing | –9.835 | 0.112 | –9.939 | –9.433 |
| $NH_3$ | Nitrogen cycle trace | –40.000 | 0.000 | –40.000 | –40.000 |

*Table II. Molecular target statistics (full 124K dataset). $H_2O$ has near-zero variance (std = 0.0002); $NH_3$ is constant at the log-floor across all samples — both are structurally problematic for supervised learning (highlighted).*

### D. Data Splitting and Preprocessing

The dataset is split 70% / 15% / 15% into train, validation, and test sets using stratified random sampling (random_state=42). All splits are fixed by index arrays, ensuring identical partitions across all experiments. At 124K samples this yields approximately 87,024 / 18,648 / 18,648 samples.

**Spectral normalisation:** Per-channel Z-score normalisation is applied to all CLIMA profiles: $Z = (x - \mu)/\sigma$, where $\mu$ and $\sigma$ have shape (12, 101) and are computed on the training set only. Any $\sigma < 10^{-8}$ is replaced by 1.0.

**PCA (RF only):** For the RF baseline, normalised profiles are flattened from (12, 101) → (1,212) and reduced to 300 principal components. These capture >90% of total variance. The PCA is fit on the training set only.

## IV. METHODS

### A. Baseline — Per-Molecule Random Forest

We train 12 independent RandomForestRegressor models (scikit-learn), one per molecule, on the 300-dimensional PCA features. To keep this model a true fast baseline, training is **capped at 10,000 samples regardless of dataset size**. Hyperparameters are tuned per molecule: molecules with high target variance ($CH_4$, CO, $H_2$) receive more estimators and deeper trees; near-constant molecules receive shallower, smaller forests.

| Molecule | n_estimators | max_depth | min_samples_leaf | max_features |
|---|---|---|---|---|
| $H_2O$, $O_3$, $CH_4$, $N_2O$, CO, $H_2$ | 300 | 16–20 | 1 | sqrt |
| $CO_2$, $O_2$ | 200 | 12 | 2 | 0.5 |
| $N_2$ | 150 | 10 | 3 | 0.4 |
| $H_2S$, $SO_2$, $NH_3$ | 400 | 20–22 | 1 | sqrt |

*Table III. Per-molecule RF hyperparameter groups. All models use n_jobs=–1, random_state=42.*

### B. Deep Model — CNN1D (Plain 1D Convolutional Network)

CNN1D is a plain (non-residual) 1D convolutional network operating directly on the Z-normalised (12, 101) CLIMA profile, with 12 per-molecule output heads. It has **368,588 trainable parameters** — reflecting that 4 convolutional layers are too shallow for residual skip connections to provide meaningful benefit, and that the task's spatial structure (altitude levels) is compressed efficiently by simple stride-2 + max-pool blocks.

**Architecture:**

| Layer / Block | Operation | Output Shape | Key Detail |
|---|---|---|---|
| Input | — | (B, 12, 101) | 12 CLIMA channels x 101 altitude levels |
| Block 1 | Conv1d(12→32, k=9, s=2, p=4) + BN + ReLU + MaxPool1d(2) | (B, 32, 25) | Large kernel captures global altitude gradients |
| Block 2 | Conv1d(32→64, k=7, s=2, p=3) + BN + ReLU + MaxPool1d(2) | (B, 64, 6) | Mid-range feature aggregation |
| Block 3 | Conv1d(64→128, k=5, s=2, p=2) + BN + ReLU + MaxPool1d(2) | (B, 128, 1) | Spatial dimension compressed to 1 |
| Block 4 | Conv1d(128→256, k=3, s=1, p=1) + BN + ReLU | (B, 256, 1) | Feature depth expansion |
| Pool | AdaptiveAvgPool1d(1) + Flatten | (B, 256) | Fixed-size representation |
| Shared FC | Dropout(0.25) + Linear(256→128) + LayerNorm(128) + ReLU | (B, 128) | Common representation for all heads |
| 12 Heads | MoleculeHead: Linear + LN + ReLU + Linear → scalar | (B, 12) | Per-molecule MLP with configurable depth |

*Table IV. CNN1D layer-by-layer architecture. No residual connections are used. BatchNorm follows each Conv1d; LayerNorm is used in the FC layers for training stability.*

Each convolutional block applies stride-2 downsampling followed by MaxPool(2), collapsing the 101-level altitude axis to 1 by the end of Block 3. Block 4 then expands channel depth from 128 to 256 at unit spatial resolution. This design avoids the projection overhead of residual shortcuts while still providing sufficient feature capacity for the task.

The 12 per-molecule heads are independently configurable in depth and dropout rate. Trace molecules with high uncertainty ($H_2S$, $SO_2$, $NH_3$) receive higher dropout (0.25); bulk molecules ($CO_2$, $O_2$, $N_2$) receive shallower heads with smaller hidden dimension (64 vs. 128). This asymmetry reduces overfitting on the near-constant molecules while preserving capacity for the high-variance biosignature molecules.

### C. Loss Function and Training

Training uses a Weighted MSE loss: $L = \mathrm{mean}(w \blacksquare (\blacksquare - y)^2)$, where w is a per-molecule importance weight vector. Weights range from 1.0 ($N_2$, bulk carrier) to 2.0 ($SO_2$, $NH_3$ — trace species that would otherwise be dominated by high-variance molecules). The optimizer is AdamW (lr = $10^{-3}$, weight decay = $10^{-4}$) with CosineAnnealingLR scheduling ($\eta$_min = $10^{-6}$). Training runs for up to 150 epochs with early stopping (patience = 30, min_delta = $10^{-5}$). Gradient clipping (max_norm = 5.0) prevents instability. During training, Gaussian noise ($\sigma = 0.01$) is added to input spectra for regularisation.

| Hyperparameter | Value | Hyperparameter | Value |
|---|---|---|---|
| Optimizer | AdamW | Batch size (train) | 32 |
| Learning rate | $1{\times}10^{-3}$ | Max epochs | 150 |
| Weight decay | $1{\times}10^{-4}$ | Early stop patience | 30 |
| LR schedule | CosineAnnealingLR | Gradient clip | 5.0 |
| $\eta$_min | $1{\times}10^{-6}$ | Augmentation | Gaussian noise $\sigma$=0.01 |
| Parameters | 368,588 | Architecture | Plain 1D CNN (no residual) |

*Table V. CNN1D training hyperparameters.*

## V. EXPERIMENTS & RESULTS

### A. Full 124K HPC Evaluation — Primary Results

The primary evaluation was performed on the Northeastern HPC cluster using the full 124,320-sample dataset. The test set contains 18,648 samples. Results are reported as $R^2$ (coefficient of determination), RMSE, and MAE — all in $\log_{10}$ units, where 1 unit = one order of magnitude. The CNN1D model was trained on the GPU partition; the RF was trained on 10,000 samples (its intentional cap) and evaluated on the full 18,648-sample test set.

| Molecule | RF $R^2$ | RF RMSE | RF MAE | CNN1D $R^2$ | CNN1D RMSE | CNN1D MAE | $\Delta R^2$ |
|---|---|---|---|---|---|---|---|
| $H_2O$ † | 0.6139 | 0.000102 | 0.0000 | 0.9788 | 0.0000240 | 0.0000 | +0.365 |
| $CO_2$ | 0.8884 | 0.2822 | 0.2084 | 0.9998 | 0.0120 | 0.0076 | +0.111 |
| $O_2$ | 0.7285 | 0.0904 | 0.0616 | 0.9987 | 0.0061 | 0.0043 | +0.270 |
| $O_3$ | 0.7520 | 0.0624 | 0.0430 | 0.9990 | 0.0040 | 0.0020 | +0.247 |
| $CH_4$ | 0.9276 | 0.4932 | 0.3832 | 0.9999 | 0.0089 | 0.0063 | +0.072 |
| $N_2$ | 0.7094 | 0.0234 | 0.0168 | 0.9979 | 0.0020 | 0.0012 | +0.289 |
| $N_2O$ | 0.7624 | 0.0748 | 0.0521 | 0.9992 | 0.0045 | 0.0032 | +0.229 |
| CO | 0.9343 | 0.3448 | 0.2819 | 0.9999 | 0.0112 | 0.0086 | +0.066 |
| $H_2$ | 0.7508 | 1.2676 | 0.9093 | 0.9984 | 0.1025 | 0.0212 | +0.248 |
| $H_2S$ | 0.9455 | 0.0932 | 0.0731 | 0.9999 | 0.0035 | 0.0023 | +0.054 |
| $SO_2$ | 0.8994 | 0.0353 | 0.0239 | 0.9998 | 0.0016 | 0.0011 | +0.100 |
| $NH_3$ ‡ | 1.0000 | 0.0000 | 0.0000 | 1.0000 | 0.0000 | 0.0000 | +0.000 |
| Mean (all 12) | 0.826 | 0.231 | 0.171 | 0.998 | 0.0130 | 0.0048 | +0.172 |

*Table VI. HPC test results — full 124K dataset, 18,648 test samples. † $H_2O$ has near-zero target variance (std ≈ 0.0002); both models achieve RMSE ≈ 0 — $R^2$ is deceptive due to near-zero denominator. ‡ $NH_3$ is constant ($\log_{10}$ = –40) across all samples; both models trivially achieve $R^2$ = 1.0. $\Delta R^2$ = CNN1D – RF; green = CNN1D wins. CNN1D outperforms RF on all 12 molecules.*

## B. Validation Results

Validation metrics (18,648 samples) confirm the test results with no signs of overfitting. CNN1D validation mean $R^2$ = 0.9993 vs. RF 0.826. The close agreement between validation and test performance indicates the model generalises well to unseen atmosphere configurations.

| Molecule | RF Val $R^2$ | CNN1D Val $R^2$ | RF Test $R^2$ | CNN1D Test $R^2$ |
|---|---|---|---|---|
| $H_2O$ | 0.597 | 0.9991 | 0.614 | 0.9788 |
| $CO_2$ | 0.887 | 0.9998 | 0.888 | 0.9998 |
| $O_2$ | 0.726 | 0.9987 | 0.729 | 0.9987 |
| $O_3$ | 0.752 | 0.9989 | 0.752 | 0.9990 |
| $CH_4$ | 0.926 | 0.9997 | 0.928 | 0.9999 |
| $N_2$ | 0.718 | 0.9982 | 0.709 | 0.9979 |
| $N_2O$ | 0.763 | 0.9992 | 0.762 | 0.9992 |
| CO | 0.933 | 0.9996 | 0.934 | 0.9999 |
| $H_2$ | 0.758 | 0.9986 | 0.751 | 0.9984 |
| $H_2S$ | 0.945 | 0.9998 | 0.945 | 0.9999 |
| $SO_2$ | 0.902 | 0.9998 | 0.899 | 0.9998 |
| $NH_3$ | 1.000 | 1.0000 | 1.000 | 1.0000 |
| Mean | 0.826 | 0.9993 | 0.826 | 0.9976 |

*Table VII. Validation vs. test $R^2$ comparison. Consistency across val/test confirms no overfitting.*

## C. Training Dynamics (CNN1D, Full 124K Run)

The CNN1D trained for the full 150-epoch budget on the HPC GPU partition without triggering early stopping. The training loss dropped from 0.0914 at epoch 1 to 0.00802 at epoch 150. The best validation loss was 0.000899 achieved at epoch 138; the final validation loss was 0.00106 (slight uptick from best due to CosineAnnealingLR nearing η_min). The absence of early stopping indicates the model continued to make marginal improvements throughout training — consistent with the very high final test accuracy.

| Metric | Epoch 1 | Epoch 150 | Best |
|---|---|---|---|
| Train loss (weighted MSE / sample) | 0.09138 | 0.00802 | — |

| | | | |
|---|---|---|---|
| Val loss (weighted MSE / sample) | 0.02727 | 0.00106 | 0.000899 (epoch 138) |
| RF mean $R^2$ (test) | — | — | 0.826 |
| CNN1D mean $R^2$ (test) | — | — | 0.998 |

*Table VIII. CNN1D training dynamics summary — 124K HPC run, 150 epochs.*

## VI. DISCUSSION

### A. H■O Anomaly

$H_2O$ shows lower $R^2$ for both models (RF: 0.614, CNN1D: 0.979) compared to other molecules. The cause is statistical: the $H_2O$ target has a standard deviation of only 0.0002 $\log_{10}$ units across the dataset — a range of just 0.003 log units total. The entire variability is below the noise floor of the CNN1D's Gaussian augmentation ($\sigma$ = 0.01). In practice, both models achieve RMSE ≈ $10^{-5}$ (essentially zero absolute error), but the near-zero denominator in $R^2 = 1 - \mathrm{MSE/Var}$ makes the metric unreliable for this molecule. This is a known limitation of $R^2$ as an evaluation metric for near-constant targets.

### B. NH■ Anomaly

$NH_3$ is at the log-floor (–40.0) for 100% of the 124,320 simulated samples — the ATMOS photochemistry model produces no detectable ammonia in this parameter space. Both the RF and CNN1D correctly learn to always predict –40.0, yielding $R^2$ = 1.0. This is a degenerate target: $R^2$ = 1.0 reflects the constant prediction matching the constant truth, not any learned atmospheric relationship. In real JWST data, $NH_3$ may vary; models should not be expected to generalise beyond the training distribution for this molecule.

### C. Biosignature Molecules

The biosignature molecules $O_3$ ($R^2$ = 0.999), $CH_4$ (0.9999), $O_2$ (0.999), and $N_2O$ (0.999) are all predicted with near-perfect accuracy by the CNN1D on the full 124K dataset. $O_3$ and $CH_4$ are particularly important — their co-detection is among the strongest evidence for biogenic chemistry [6]. The CNN1D's near-perfect recovery of both suggests it has learned the underlying photochemical relationships that link these molecules to the atmospheric state encoded in the CLIMA profile.

### D. RF Performance at 124K Scale

The RF mean $R^2$ of 0.826 on the 18,648-sample test set is lower than might be expected. This is entirely a consequence of the intentional 10,000-sample training cap: the RF is evaluated on an 18,648-sample test set drawn from the full 124K parameter space, while it was trained on only a 10K subset. The RF's decision tree structure cannot interpolate in the tails of the distribution as effectively as the CNN1D, which trained on 87,024 samples covering the full parameter space. The RF remains a useful and fast baseline — achieving $R^2$ > 0.9 on the well-represented high-variance molecules ($CH_4$: 0.928, CO: 0.934, $H_2S$: 0.945) — but is clearly outclassed by the neural network.

### E. Computational Efficiency

Both models perform inference in milliseconds per planet once trained, compared to hours for Bayesian nested sampling. The CNN1D processes a batch of 64 CLIMA profiles in ~2 ms on Apple Silicon MPS (single forward pass). The RF inference time is similarly fast (~1 ms for 12 parallel forest evaluations). This ~10,000× speedup over Bayesian methods makes population-level surveys (thousands of planets) computationally feasible.

### F. Limitations and Future Work

- **Synthetic data only:** Models are trained on ATMOS simulations. Real JWST spectra differ in noise characteristics, instrument resolution, and systematic biases. Validation on JWST observations (e.g., WASP-39 b) is a critical next step.
- **No uncertainty quantification:** The current models produce point estimates only. Bayesian neural networks or MC Dropout could provide posterior distributions comparable to nested sampling.
- **PSG spectra:** A complementary dataset of 0.2–2.0 µm transmission spectra (4,378 wavelength points) is available from the same archive but not yet incorporated. Training on actual spectra would be more directly applicable to JWST observations.

- **$H_2O$ and $NH_3$:** These targets require either dataset curation (filtering constant-target samples) or model redesign (dedicated constant-detection logic).
- **Further scaling:** The full INARA archive contains ~3.1 million samples. Given the clear data-scaling benefit observed in this work, further scaling is a direct path to improved retrieval accuracy.

## VII. CONCLUSION

We trained and evaluated two ML models for exoplanet atmospheric retrieval on the INARA ATMOS benchmark dataset. The plain 1D CNN (CNN1D, 368,588 parameters) and per-molecule Random Forest both achieve scientifically useful accuracy at millisecond inference time, representing a ~10,000× speedup over Bayesian retrieval methods.

On the full 124K HPC run, the CNN1D achieves mean $R^2 = 0.998$ across all 12 molecules on an 18,648-sample test set, outperforming the RF (mean $R^2 = 0.826$) on every molecule. The CNN1D's near-perfect recovery of biosignature molecules — $O_3$, $CH_4$, $O_2$, $N_2O$ all at $R^2 > 0.998$ — demonstrates that the CLIMA profile contains sufficient information to determine the full molecular composition with exceptional accuracy.

The gap between the two models reflects the difference in training data: the RF is intentionally capped at 10,000 samples regardless of dataset size, while the CNN1D trains on the full 87,024-sample training split. This design choice keeps the RF a true lightweight baseline — and the CNN1D's compact architecture (368,588 parameters, no residual connections) demonstrates that plain convolutional networks are sufficient for this retrieval task when sufficient training data is available.

These results validate the INARA benchmark as a useful test bed for ML retrieval algorithms and demonstrate that data scale — not model capacity — is the primary bottleneck. Future work should focus on validation against real JWST spectra, uncertainty quantification, and incorporation of the PSG spectral dataset for direct comparison with Bayesian retrieval posteriors.

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